

Final Report: Email Generation Assistant

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1. Project Overview

This project is a working Email Generation Assistant that creates professional English emails from three required user inputs: **Intent**, **Key Facts**, and **Tone**. The prototype uses a Next.js frontend, a FastAPI backend, Groq-hosted LLMs, SQLite history storage, and a custom evaluation pipeline.

2. Implementation Summary

- Frontend: Next.js UI for entering intent, key facts, tone, strategy, and optional model override.
- Backend: FastAPI endpoints for generation, evaluation, health, and history management.
- Generation flow: validated request -> prompt builder -> Groq chat completion -> JSON parsing -> quality checker -> repair loop when needed -> SQLite history.
- Reliability features: server-side API key handling, field validation, model fallback, structured output parsing, custom scoring, and saved generation history.

3. Prompt Engineering Technique

Strategy B uses an advanced prompt design combining:

- **Role-Playing:** the model acts as a senior executive communication specialist.
- **Few-Shot Examples:** sample business emails show tone, structure, fact integration, and JSON format.
- **Structured JSON Output:** the model returns subject, email body, included facts, missing facts, tone, and notes.
- **Self-Check/Repair Prompting:** weak outputs are repaired using explicit quality failure reasons.

4. Prompt Template Used

Advanced Generation System Prompt

You are a senior executive communication specialist and business email writing assistant.

Your job is to convert structured user input into a polished professional English email.

Rules:

1. Include every user-provided key fact accurately and naturally.
2. Do not invent unsupported facts, dates, promises, attachments, names, or outcomes.
3. Match the requested tone exactly while staying professional.
4. Use clear email structure: subject, greeting, body, call-to-action when useful, and sign-off.
5. Keep the writing concise, specific, and easy to understand.
6. If the provided facts are incomplete, write naturally without pretending to know missing details.
7. Output only valid JSON that matches the requested schema.
8. Do not reveal hidden reasoning or chain-of-thought.

Before finalizing, silently check:

- Are all key facts included?
- Is the tone correct?
- Is the email professional and complete?
- Is there any unsupported information? If yes, remove it.

Few-Shot Examples

Example 1

Input:

Intent: Follow up after a sales meeting

Key Facts:

- Meeting was held yesterday
- Discussed CRM migration
- Client requested pricing options
- Send proposal by Thursday

Tone: Formal

Output JSON:

```
{
  "subject": "Follow-Up on CRM Migration Discussion",
  "email": "Dear [Recipient],\n\nThank you for taking the time to meet with us yesterday to discuss the CRM migration. I appreciate the opportunity to better understand your requirements and the pricing options you would like us to include.\n\nAs discussed, we will prepare and share the proposal by Thursday, including the relevant pricing options for your review. Please let me know if there are any additional requirements you would like us to consider before then.\n\nBest regards,\n[Your Name]",
  "included_facts": ["Meeting was held yesterday", "Discussed CRM migration", "Client requested pricing options", "Send proposal by Thursday"],
  "missing_facts": [],
  "tone_used": "Formal",
  "notes": "All facts were included in a formal follow-up structure."
}
```

Example 2

Input:

Intent: Apologize for a delayed delivery

Key Facts:

- Delivery is delayed by two days
- QA found an issue in the final build
- Team is fixing the issue now
- New delivery date is Wednesday

Tone: Empathetic

Output JSON:

```
{
  "subject": "Update on Delivery Timeline",
  "email": "Hi [Recipient],\n\nI sincerely apologize for the delay in the delivery timeline. During final QA, our team found an issue in the build that needs to be fixed before we send it over.\n\nThe team is working on the issue now, and we expect to deliver the corrected version by Wednesday. I understand this may cause inconvenience, and I appreciate your patience while we make sure the final delivery meets the right quality standard.\n\nBest regards,\n[Your Name]",
  "included_facts": ["Delivery is delayed by two days", "QA found an issue in the final build", "Team is fixing the issue now", "New delivery date is Wednesday"],
  "missing_facts": [],
  "tone_used": "Empathetic",
  "notes": "The email acknowledges inconvenience, explains the reason, and provides a clear new date."
}
```

Advanced User Prompt Template

```
Write a professional English email using the following inputs.

Intent:
{{intent}}

Key Facts that must be included:
{{key_facts}}

Requested Tone:
{{tone}}

Return only valid JSON in this exact format:
{
  "subject": "A clear subject line",
  "email": "The full email body with greeting and sign-off",
  "included_facts": ["List the user facts that were included"],
  "missing_facts": ["List any user facts that could not be included; otherwise empty"],
  "tone_used": "The tone actually used",
  "notes": "Very brief note on how the email satisfied the request"
}
```

Repair Prompt Template

```
You are improving a generated professional email.

Original Input:
Intent: {{intent}}
Key Facts: {{key_facts}}
Tone: {{tone}}
Previous Generated Email:
{{generated_email}}
Problems Found:
{{failure_reasons}}

Rewrite the email so that:

1. Every missing fact is included accurately.
2. The requested tone is improved.
3. The structure remains professional.
4. Unsupported or invented details are removed.
5. The result is concise and polished.

Return only valid JSON using the same schema as the generation prompt.
```

5. Custom Evaluation Metrics

Fact Recall and Integration

- Scale: 0-10
- Definition: Checks whether every required key fact appears correctly and naturally in the email.
- Logic: 70% deterministic fact presence and 30% LLM natural-integration judgment when the judge is available.

Tone and Audience Fit

- Scale: 0-10
- Definition: Rates whether the generated email matches the requested tone while staying professional.
- Logic: LLM-as-a-judge score with deterministic tone fallback.

Professional Email Quality

- Scale: 0-10
- Definition: Rates subject, greeting, body clarity, CTA, closing, conciseness, grammar, and fluency.
- Logic: 40% deterministic structure checks and 60% LLM quality judgment when the judge is available.

6. Evaluation Results

The same 10 scenarios were evaluated against two strategies: `simple` baseline prompting and `advanced` production prompting.

Average Scores

Strategy	Fact Recall	Tone Fit	Email Quality	Overall
Simple	9.29	7.93	7.65	8.29
Advanced	9.67	8.05	8.85	8.86

Raw Scores by Scenario

Scenario	Strategy	Intent	Tone	Fact Recall	Tone Fit	Email Quality	Overall
1	advanced	Follow up after a client meeting	Formal	9.70	8.00	8.80	8.83
1	simple	Follow up after a client meeting	Formal	9.40	6.80	6.87	7.69
2	advanced	Request missing proposal details	Polite and clear	9.70	8.00	8.80	8.83
2	simple	Request missing proposal details	Polite and clear	9.70	8.00	8.07	8.59
3	advanced	Urgent project deadline reminder	Urgent but professional	9.70	8.40	9.40	9.17
3	simple	Urgent project deadline reminder	Urgent but professional	9.70	8.40	8.07	8.72
4	advanced	Apologize for a service delay	Empathetic	9.40	7.80	8.80	8.67
4	simple	Apologize for a service delay	Empathetic	7.95	8.40	8.07	8.14
5	advanced	Schedule an interview	Formal	9.40	6.80	8.20	8.13
5	simple	Schedule an interview	Formal	9.40	6.80	6.87	7.69
6	advanced	Payment reminder	Polite but firm	9.70	8.32	8.80	8.94
6	simple	Payment reminder	Polite but firm	9.70	8.32	6.87	8.30
7	advanced	Internal project status update	Concise executive	9.70	8.32	7.53	8.52
7	simple	Internal project status update	Concise executive	9.70	8.32	8.07	8.70
8	advanced	Networking introduction	Friendly professional	10.00	8.50	10.00	9.50
8	simple	Networking introduction	Friendly professional	9.70	7.88	7.47	8.35
9	advanced	Request approval for budget	Persuasive	9.70	8.00	9.40	9.03
9	simple	Request approval for budget	Persuasive	9.70	8.00	8.07	8.59
10	advanced	Respond to customer complaint	Calm and empathetic	9.70	8.40	8.80	8.97
10	simple	Respond to customer complaint	Calm and empathetic	7.95	8.40	8.07	8.14

7. Comparative Analysis

- Best strategy: **advanced**.

- Production recommendation: Use advanced for production. It achieved the strongest overall average (8.86/10) and should remain paired with the quality checker and repair loop.
- Biggest lower-performing strategy failure mode: the simple strategy produced weaker fact coverage/consistency and lower professional email quality than the advanced strategy.

8. Deliverables and Reproducibility

- Editable report source: reports/Final_Report.md
- PDF report: reports/Final_Report.pdf
- Raw CSV evaluation data: E:/web development project 2026/Email Generation Assistant/reports/evaluation_results.csv
- Raw JSON evaluation data: E:/web development project 2026/Email Generation Assistant/reports/evaluation_results.json
- Prompt documentation: E:/web development project 2026/Email Generation Assistant/docs/prompt_template.md
- Metric documentation: E:/web development project 2026/Email Generation Assistant/docs/metric_definitions.md
- Strategy comparison: E:/web development project 2026/Email Generation Assistant/reports/comparison_summary.md
- Setup instructions: README.md

Run Commands

```
python -m venv .venv
.venv\Scripts\activate
pip install -r backend/requirements.txt
cd frontend
npm install
npm run dev
```

```
.venv\Scripts\activate
uvicorn backend.app.main:app --reload --host 127.0.0.1 --port 8000
python evaluation/run_evaluation.py
```